Quantitative Aptitude

BankPrep AI | Banking Exam Preparation Study Material
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1. Percentages

Percentage means a value out of 100. Percentage = (Part / Whole) x 100.

Example: 20% of 250 = (20 / 100) x 250 = 50.

Question: Find 15% of 400. Answer: 60.

2. Ratio and Proportion

A ratio compares quantities. For example, 2:3 means two parts compared with three parts.

Example: Divide 500 in the ratio 2:3. Total parts = 5. Shares = 200 and 300.

Question: Divide 720 in the ratio 5:7. Answer: 300 and 420.

3. Profit and Loss

Profit = Selling Price - Cost Price.

Loss = Cost Price - Selling Price.

Profit percentage = (Profit / Cost Price) x 100.

Question: An item costs Rs. 800 and sells for Rs. 920. Answer: Profit = Rs. 120; profit percentage = 15%.

4. Simple and Compound Interest

Simple Interest = (Principal x Rate x Time) / 100.

Example: Principal Rs. 1,000, annual rate 5%, time 2 years. Simple interest = Rs. 100.

Compound interest is calculated on the principal plus previously accumulated interest.

Question: Find simple interest on Rs. 2,000 at 6% per year for 3 years. Answer: Rs. 360.

5. Averages

Average = Sum of observations / Number of observations.

Question: Find the average of 10, 20, 30, 40 and 50. Answer: 30.

6. Speed, Time and Distance

Speed = Distance / Time.

Question: A vehicle travels 150 km in 3 hours. Answer: Average speed = 50 km/h.

7. What Changed in Quantitative Aptitude Questions (2026)

The syllabus for banking quantitative aptitude is largely the same, but the way questions are asked has shifted. Practice questions in this file are original, written in the style of questions reported in recent exams. They are not actual exam questions.

SBI PO Prelims 2026 (1 and 2 August): candidates reported arithmetic questions on boats, ages, trains, mensuration, averages, profit and loss, partnership and time and work, plus missing number series and mixtures. In at least one shift no approximation or simplification question appeared.

IBPS PO Prelims 2026 (22 and 23 August): Data Interpretation (DI) was reported as simple one-liner sets, line graph DI and table DI. In one shift there was no simplification, missing number series or quadratic equation question.

SBI Clerk Prelims 2026 (26 and 27 September): Numerical Ability stressed speed mathematics and direct DI (including caselet DI), simplification, number series and quadratic equations, with arithmetic asked as direct single-concept questions.

SBI PO Mains 2026 and IBPS PO Mains 2026: Data Analysis and Interpretation was the toughest, most calculation-heavy section. Reported content: several DI sets, arithmetic word problems, number series, quadratic equations and data sufficiency.

Practice implication: expect fewer pure formula questions and more set-based and comparison-style questions. Learn to compare quantities by estimating rather than calculating fully.

8. Quantity I versus Quantity II Questions

In this format two quantities are given and you decide how they relate. Often you do not need exact values. Standard options for comparison questions: (a) Quantity I > Quantity II; (b) Quantity I < Quantity II; (c) Quantity I >= Quantity II; (d) Quantity I <= Quantity II; (e) Quantity I = Quantity II or no relation can be established.

Q1. Quantity I: speed of a boat in still water if its downstream speed is 15 km/h and upstream speed is 9 km/h. Quantity II: 10 km/h. Answer: Quantity I = (15 + 9) / 2 = 12 km/h, so Quantity I > Quantity II (option a).

Q2. Quantity I: 30% of 450. Quantity II: 45% of 300. Answer: Both are 135, so Quantity I = Quantity II (option e).

Q3. Quantity I: simple interest on Rs. 4,000 at 5% per year for 3 years. Quantity II: compound interest on Rs. 2,000 at 10% per year for 2 years. Answer: Quantity I = Rs. 600; Quantity II = 2,000 x (1.21 - 1) = Rs. 420. Quantity I > Quantity II (option a).

9. Data Sufficiency in Quantitative Aptitude

Data sufficiency questions ask whether the given statements are enough to answer a question. You do not need the final value unless asked. Standard options: (a) Statement I alone is sufficient; (b) Statement II alone is sufficient; (c) either statement alone is sufficient; (d) both statements together are needed; (e) even both together are not sufficient.

Q1. What is the present age of A? I. Five years ago A was twice as old as B. II. B is 15 years old now. Answer: (d). From II, B five years ago was 10, so A five years ago was 20 and A is 25 now. Neither statement alone gives A's age.

Q2. What is the area of a rectangle? I. The perimeter is 40 cm. II. The length is 4 cm more than the breadth. Answer: (d). Together, l + b = 20 and l - b = 4, so l = 12 and b = 8 and the area is 96 square cm.

10. Missing Number and Wrong Number Series

Series patterns now mix operations. Look at differences first, then multiply-and-add patterns, then squares and cubes.

Q1. Find the missing number: 7, 9, 13, 21, 37, ?. Answer: 69. Differences are 2, 4, 8, 16, so the next difference is 32 and 37 + 32 = 69.

Q2. Find the missing number: 3, 4, 10, 33, 136, ?. Answer: 685. The pattern is x1 + 1, x2 + 2, x3 + 3, x4 + 4, x5 + 5: 136 x 5 + 5 = 685.

Q3. Find the wrong number: 5, 11, 23, 47, 96, 191, 383. Answer: 96. Each term is the previous term x 2 + 1, so 47 should be followed by 95.

11. Quadratic Equations: Compare x and y

Solve both equations to get the possible values of x and y, then compare every pair. Use the standard options: x > y, x < y, x >= y, x <= y, or x = y or no relation.

Q1. I. x^2 - 7x + 12 = 0. II. y^2 - 9y + 20 = 0. Answer: x = 3 or 4; y = 4 or 5. Every x is less than or equal to every y, so x <= y.

Q2. I. 2x^2 + 11x + 15 = 0. II. y^2 + 7y + 12 = 0. Answer: x = -3 or -2.5; y = -4 or -3. Every x is greater than or equal to every y, so x >= y.

Q3. I. x^2 - 5x + 6 = 0. II. y^2 - 7y + 10 = 0. Answer: x = 2 or 3; y = 2 or 5. Comparisons go both ways (3 > 2 but 2 < 5), so no relation can be established.

12. Arithmetic Word Problems in 2026 Style

Boats and streams. Q1. A boat goes 90 km downstream and 90 km back. Its speed in still water is 12 km/h and the stream speed is 3 km/h. Find the total time. Answer: Downstream speed 15 and upstream speed 9. Time = 90/15 + 90/9 = 6 + 10 = 16 hours.

Ages. Q2. The present ages of A and B are in the ratio 4:5. After 6 years the ratio will be 5:6. Find A's present age. Answer: (4k + 6)/(5k + 6) = 5/6 gives k = 6, so A = 24 years (B = 30).

Trains. Q3. A 150 m long train crosses a pole in 10 seconds. How long will it take to cross a 300 m long platform? Answer: Speed = 15 m/s (54 km/h). Time = (150 + 300) / 15 = 30 seconds.

Mensuration. Q4. A cylinder has radius 7 cm and height 10 cm. Find its curved surface area and volume (pi = 22/7). Answer: Curved surface area = 2 x pi x r x h = 440 square cm. Volume = pi x r^2 x h = 1,540 cubic cm.

Partnership. Q5. A invests Rs. 25,000 for 12 months, B invests Rs. 30,000 for 8 months and C invests Rs. 20,000 for 6 months. Profit is Rs. 55,000. Find each share. Answer: Ratio of capital x time = 300,000 : 240,000 : 120,000 = 5:4:2. Shares: A Rs. 25,000, B Rs. 20,000, C Rs. 10,000.

Mixture. Q6. A 40 litre mixture has milk and water in the ratio 3:1. How much water must be added to make the ratio 3:2? Answer: Milk = 30 L, water = 10 L. For 3:2 water must be 20 L, so add 10 litres.

Time and work. Q7. A can finish a job in 12 days and B in 18 days. How long will they take together? Answer: Combined rate = 1/12 + 1/18 = 5/36 per day, so 36/5 = 7.2 days.

Profit and loss with discount. Q8. An item is marked at Rs. 800 and sold at 15% discount. The cost price is Rs. 500. Find the profit percentage. Answer: Selling price = Rs. 680; profit = Rs. 180; profit % = 180/500 x 100 = 36%.

Average. Q9. The average of 5 numbers is 24. If one number, 30, is removed, what is the average of the remaining four? Answer: Total = 120; remaining total = 90; average = 22.5.

Probability. Q10. A bag has 5 red and 3 blue balls. Two balls are drawn without replacement. Find the probability that both are red. Answer: 5C2 / 8C2 = 10/28 = 5/14.

Compound interest. Q11. Find the compound interest on Rs. 8,000 at 10% per year for 2 years. Answer: 8,000 x (1.1^2 - 1) = Rs. 1,680.

13. Data Interpretation Practice

One-liner DI. Q1. In a class of 60 students, 45% are girls. How many are boys? Answer: 55% of 60 = 33 boys.

Table DI. Production of four companies in thousand units. Company A: 2023 = 40, 2024 = 50. Company B: 2023 = 60, 2024 = 54. Company C: 2023 = 45, 2024 = 63. Company D: 2023 = 35, 2024 = 42.

Q2 (Table DI). What is the percentage increase in production of Company A from 2023 to 2024? Answer: (50 - 40)/40 x 100 = 25%.

Q3 (Table DI). Which company had the highest percentage change from 2023 to 2024? Answer: Company C. Changes are A +25%, B -10%, C +40%, D +20%.

Q4 (Table DI). By what percentage did total production increase from 2023 to 2024? Answer: Total 2023 = 180 and total 2024 = 209. Increase = 29/180 x 100 = approximately 16.1%.

Line graph DI. Daily sales of a shop (units) from Monday to Friday: 120, 150, 90, 180, 160.

Q5 (Line graph). Find the average daily sales. Answer: 700/5 = 140 units. Q6. What is the percentage increase in sales from Wednesday to Thursday? Answer: (180 - 90)/90 x 100 = 100%.

Caselet DI. A school has 600 students, of whom 40% are girls. 25% of the boys and 30% of the girls play cricket.

Q7 (Caselet). How many students play cricket in total? Answer: Girls = 240 and boys = 360. Boys playing cricket = 90 and girls playing cricket = 72. Total = 162, which is 27% of all students. Q8. How many boys do not play cricket? Answer: 360 - 90 = 270.

14. Approximation and Simplification

These appear less often in recent PO Prelims but remain in clerk-level exams. Round numbers sensibly, then calculate.

Q1. Approximate value: 24.98% of 2000.1 + 14.97 x 10.02. Answer: About 25% of 2000 + 15 x 10 = 500 + 150 = 650.

Q2. Simplify: (18 x 15 / 9) + (25^2 / 5) - 12. Answer: 30 + 125 - 12 = 143.

Q3. Simplify: 15% of 480 + 40% of 350. Answer: 72 + 140 = 212.